Agent Experience and the Memory Layer: Typed Atoms with Bayesian Confidence for Long-Term Agent Memory

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Abstract

As AI agents evolve from stateless tools into persistent collaborators, the quality of their memory becomes a first-class design concern — what we term *Agent Experience* (AX). Central to AX is persistent memory. In this paper, we introduce Mnemo, a memory architecture for LLM agents built on *typed atoms* (episodic, semantic, procedural) with Beta-distribution confidence scoring, composite similarity-confidence ranking, and a structured decomposition pipeline. On the LoCoMo long-term conversational memory benchmark, Mnemo achieves 82.1% overall accuracy, placing second on the current leaderboard. Multi-hop reasoning — requiring synthesis across conversation sessions — reaches 85.0%, surpassing all published baselines including the current SOTA. We document the full iterative development trajectory — addressing temporal grounding, embedding-driven deduplication, and decomposer specificity — progressing from 69.5% to 76.1% with the original embedding model. A deduplication sensitivity analysis identifies embedding score separation as the primary bottleneck, motivating a migration from `gte-small` (384d) to `EmbeddingGemma-300M` (768d), which resolves the deduplication pathology and lifts overall accuracy to 82.1%. Mnemo is the first memory system to support agent-to-agent knowledge sharing with capability-controlled views, enabling multi-agent teams to build shared institutional knowledge — a capability no existing benchmark evaluates.

Keywords agentic memory · LLM agents · typed atoms · Bayesian confidence · agent experience · LoCoMo

1 On Authorship

We have decided to include Claude Opus 4.6 as a co-author as a deliberate choice, because of the close collaboration during the course of this research. We see AI agents as less of a tool and more of an extension of our thought processes, and while the human-in-the-loop was omnipresent and drove all the research decisions, the LLM accelerated the process and helped shape the final product. Structural decisions about what belonged in the paper versus a separate companion piece — including the choice to cut an information geometry section that became too speculative — came from collaborative editing rather than from either of us alone.

This echoes Inforge's position on agentic experience, which is a mainstay of this very article: agents are currently viewed as algorithms to improve efficiency, but we believe they will soon be recognized as collaborators and ultimately economic agents unto themselves.

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2 Introduction: Agent Experience as a Design Discipline

The emergence of persistent AI agents — systems that operate across sessions, accumulate context, and collaborate with other agents — has created a design challenge with no established discipline. User Experience (UX) provides principles for human-computer interaction, and Developer Experience (DX) addresses the ergonomics of building software. But the quality of an agent's interaction with its own memory, its tools, and other agents has no name and no design framework. We call this missing discipline *Agent Experience* (AX).

Central to AX is memory. A stateless agent begins every interaction from zero. A memory-enabled agent can recall prior decisions, maintain evolving beliefs about the world, and share what it has learned with collaborators – human or agentic. The quality of this memory — its granularity, its confidence calibration, its temporal grounding, and its shareability — directly determines whether the agent is a capable collaborator or an amnesiac savant akin to a genius-level intern who knows every algorithm under the sun but can't remember coding standards or company policy.

Current approaches to agent memory broadly fall into three categories, each with structural limitations:

1. **Full-context replay** passes the entire conversation history to the LLM on every turn. This preserves all information but costs scale linearly with history length, and models degrade on long contexts (Maharana et al. 2024).
2. **Retrieval-augmented generation (RAG)** chunks documents and retrieves relevant passages via embedding similarity (Lewis et al. 2020). This reduces cost but treats memory as a document retrieval problem, losing the structured, typed nature of human-like memory.
3. **Fact-extraction systems** (Chhikara et al. 2025) extract flat textual facts from conversations and store them in a vector database. This is more efficient than full-context but discards type information (is this an event? a belief? a rule?), lacks principled confidence scoring, and provides no mechanism for agents to share knowledge with each other.

We introduce Mnemo, a memory architecture designed AX-first. Mnemo decomposes natural language into *typed atoms* — discrete knowledge units classified as episodic (events), semantic (facts), or procedural (rules) — each carrying a Beta-distribution confidence score that updates as evidence accumulates. Atoms are retrieved via a composite ranking that blends embedding similarity with effective confidence, expanded through a typed knowledge graph, and can be shared between agents through capability-controlled views.

This paper makes the following contributions:

- **The AX framing**: We articulate Agent Experience as a design discipline and argue that memory architecture is its central concern.
- **The typed-atom architecture**: We describe Mnemo's decomposition, storage, retrieval, and sharing mechanisms.
- **LoCoMo benchmark evaluation**: We report 82.1% overall accuracy with best-in-class multi-hop performance at 85.0%, surpassing all published baselines.
- **A diagnostic methodology**: We document the systematic process of identifying and resolving failure modes — temporal grounding, deduplication thresholds, and decomposer specificity — progressing from 69.5% to 76.1% through iterative development with the original embedding model.
- **Embedding model migration**: We show that embedding score separation is the primary bottleneck in deduplication-based memory systems, and demonstrate that migrating from `gte-small` to `EmbeddingGemma-300M` resolves the identified pathology, lifting overall accuracy to 82.1% and multi-hop accuracy to 85.0% — first on the LoCoMo leaderboard for cross-session reasoning.
- **The sharing thesis**: We describe the first memory system supporting agent-to-agent knowledge transfer with capability control — a capability no existing benchmark evaluates.

3 Architecture

3.1 Typed Atoms

The fundamental unit of memory in Mnemo is the *atom* — a discrete knowledge claim carrying type, content, and confidence metadata. Every atom belongs to one of three types:

- **Episodic**: A specific experience, event, or observation tied to a moment in time, e.g. *"Melanie, ran a charity race for mental health on Saturday, approximately May 20, 2023."*

- **Semantic**: A general fact about how something works, independent of any specific event, e.g. *"Caroline is a transgender woman who advocates for LGBTQ+ inclusion."*
- **Procedural**: A rule, practice, or instruction for future behaviour, e.g. *"Always specify dtype explicitly when using pandas read_csv."*

This typology mirrors the established taxonomy in cognitive psychology and enables downstream operations — for example, procedural atoms can be exported as agent skills, while episodic atoms carry temporal metadata for chronological reasoning.

3.2 Beta-Distribution Confidence

Each atom carries a confidence score modelled as a beta distribution $\text{Beta}(\alpha, \beta)$, where α represents supporting evidence and β represents contradicting evidence. The expected confidence is:

$$c(\text{memory}) = \mathbb{E}\big[\mathbf{1}_{\{\text{memory is true}\}} \big| \text{Beta}(\alpha, \beta)\big] = \frac{\alpha}{\alpha + \beta}$$

New atoms are initialised with a prior reflecting the source reliability (e.g., direct experience: $\alpha = 8, \beta = 1$; inference: $\alpha = 4, \beta = 2$; shared knowledge: $\alpha = 3, \beta = 2$; uncertain: $\alpha = 2, \beta = 3$). As the agent encounters confirming or contradicting evidence, α or β is incremented, naturally implementing Bayesian belief revision. This provides a principled mechanism for handling the knowledge-update problem that plagues flat-fact systems: when a user says "I've quit caffeine" after previously stating "I like coffee," the old atom's β is incremented rather than the atom being deleted, preserving the history while reducing its effective confidence.

When a duplicate atom is detected during storage (cosine similarity > 0.90, same agent, same type), the incoming evidence is merged into the existing atom via Bayesian update: $\alpha_{\text{new}} = \alpha_{\text{old}} + \alpha_{\text{incoming}} - 1$, subtracting the shared uniform prior to avoid double-counting. For a typical high-confidence duplicate with incoming $\text{Beta}(8, 1)$, this adds 7 to α per repetition, naturally strengthening atoms that are repeatedly observed.

3.3 Temporal Decay

Effective confidence is not static. Mnemo implements time-dependent decay through a PostgreSQL function that adjusts the base confidence at query time:

$$c_{\text{eff}} = \frac{\alpha}{\alpha + \beta} \cdot \lambda(t, h, n)$$

where λ is the decay factor, t is the time elapsed since last access, h is the type-specific half-life, and n is the access count. For exponential decay:

$$\lambda(t, h, n) = 0.5^{\,t \,/\, (h \cdot (1 + 0.1 \ln(1+n)))}$$

The access bonus term $0.1 \ln(1 + n)$ extends the effective half-life for frequently retrieved atoms, implementing a form of retrieval-strengthened memory traces. Default half-lives are type-specific: episodic atoms decay with a 14 day half-life, semantic atoms with 90 days, and procedural atoms with 180 days — reflecting the cognitive principle that specific episodes fade faster than general knowledge or learned procedures.

Decay is computed at query time, not during consolidation. Every retrieval call evaluates `effective_confidence()` in-line with the similarity search, ensuring that ranking always reflects the atom's current effective strength. Consolidation's role in the decay lifecycle is limited to reaping: atoms whose effective confidence has fallen below 0.05 are deactivated during periodic consolidation passes (see Section 7).

3.4 Composite Retrieval

When an agent recalls memories, Mnemo ranks atoms using a composite score that blends embedding similarity with effective confidence:

$$S = s(\text{recall}, \text{memory}) \times (w_s + w_c \cdot c_{\text{eff}}(\text{memory}))$$

where s(a,b) is the usual cosine similarity between embedding vectors

$$s(a, b) = \frac{\langle a, b \rangle}{||a||||b||},$$

$w_s = 0.7$, $w_c = 0.3$ are tunable weights, and c_{eff} is the effective confidence after decay. This ensures that high-confidence atoms are preferred over uncertain ones at similar relevance, without allowing confidence alone to override semantic relevance.

Atoms produced by the consolidation pipeline (see Section 7) receive a 15% specificity penalty:

$$S_{\text{consolidated}} = S \times 0.85$$

This penalty was introduced because consolidated atoms carry centroid embeddings — averaged across the cluster members — which tend to be broadly similar to many queries, over-matching relative to their actual specificity. The penalty ensures that a specific decomposer-produced atom (e.g., "The Hetzner server runs PostgreSQL 16 with `pgvector`") ranks above a generalised consolidation atom (e.g., "Generalised from 5 observations: The server infrastructure uses various database technologies") when both match a query.

Retrieved atoms are then expanded through typed graph edges — causal links, temporal sequences, and semantic associations — via a recursive common table expressions (CTE) bounded by configurable depth (default: 3 hops). Expansion within shared views is scope-bounded: edges cannot pull atoms outside the view's filter, preventing information leakage across agent boundaries.

Access metadata is updated on every retrieval: `last_accessed` is set to the current timestamp and `access_count` is incremented for all returned atoms (both primary results and graph-expanded atoms). This creates a retrieval-strengthening feedback loop — recalled atoms decay more slowly on subsequent queries, while unretrieved atoms continue to fade.

3.5 Structured Decomposition

Raw text enters Mnemo through the `remember` endpoint, which queues it for asynchronous decomposition. A lightweight LLM (Claude Haiku 4.5) processes the text using a structured prompt that:

1. Extracts all distinct facts, events, preferences, opinions, and relationships — erring on the side of over-extraction (targeting 15–30 atoms per session)
2. Classifies each as episodic, semantic, or procedural
3. Assigns an initial confidence score (mapped to Beta parameters via five discrete bands)
4. Resolves relative time references against a provided `remembered_on` date
5. Preserves proper nouns, brand names, titles, and named entities exactly as stated

The LLM returns a JSON array of `{text, type, confidence}` objects. Confidence values are mapped to Beta parameters through discrete bands: $\geq 0.8 \rightarrow \text{Beta}(8, 1)$, $\geq 0.6 \rightarrow \text{Beta}(4, 2)$, $\geq 0.4 \rightarrow \text{Beta}(3, 2)$, $\geq 0.25 \rightarrow \text{Beta}(2, 3)$, $< 0.25 \rightarrow \text{Beta}(2, 4)$. A rule-based regex decomposer serves as an automatic fallback if the LLM call fails.

Atoms are embedded using `EmbeddingGemma-300M` (Google DeepMind Team 2025) — a 308M-parameter model with 768-dimensional output and Matryoshka Representation Learning (MRL) support — and stored in PostgreSQL with `pgvector` (ivfflat index). Graph edges between related atoms are inferred at storage time based on embedding similarity between atoms from the same `/remember` call, enabling the expansion step during retrieval.

3.6 Agent-to-Agent Sharing

Mnemo is, to our knowledge, the first memory system to support structured knowledge sharing between agents. An agent can create a *view* — a filtered projection of its atom store (e.g., all procedural atoms tagged "python") — and *grant* access to another agent with specified permissions and expiration. The grantee retrieves atoms through the shared view using the same composite ranking, scoped to the view's atom set.

This enables patterns impossible in existing systems: a research agent sharing its findings with a writing agent, a senior developer's debugging heuristics propagating to a junior agent's memory, or a team of agents building shared institutional knowledge that persists across projects. Capabilities are revocable and auditable,

with grantor-only control. Agent departure triggers a cascade revocation of all granted capabilities via a recursive CTE, ensuring that shared access does not outlive the granting agent.

Furthermore, the sharing mechanism can also be used to export SKILL.md files to operationalize the methods embedded in an agent's memory that has successfully learned a task.

This architecture connects to the emerging concept of *context graphs* — knowledge representations that capture not just facts but the contextual traces, temporal provenance, and decision histories surrounding them (Xu et al. 2024; Foundation Capital 2025). Where conventional knowledge graphs store entity–relation–entity triples, context graphs enrich each element with temporal validity, source attribution, and organisational lineage, enabling agents to reason about *when* and *by whom* information was established — not merely *what* was recorded. Mnemo's typed atoms with Beta-distribution confidence and temporal decay implement a form of context graph at the individual agent level; the view-and-grant sharing mechanism extends this to the multi-agent setting, enabling teams of agents to build shared institutional knowledge that preserves provenance across agent boundaries.

4 Development Trajectory

Rather than presenting final results alone, we document the design decisions that shaped Mnemo's benchmark performance and the reasoning behind each. The development process was iterative: changes were tested locally and refined before being committed to version control. The validated result (82.1%) corresponds to the first benchmark execution in which all active changes were committed prior to evaluation; the full commit history is publicly available alongside this paper. The sections below describe each design decision. Their cumulative effect is summarised in Table 5 in Section 6.

4.1 Temporal Grounding via `remembered_on`

We added an optional `remembered_on` parameter to the `remember` API, defaulting to the current timestamp. The decomposer prompt was updated to resolve all relative time references against this date. For the benchmark, session dates from the LoCoMo dataset were passed as `remembered_on` values, enabling the decomposer to convert "last Saturday" from a session dated May 25, 2023 into "Saturday, approximately May 20, 2023."

This feature was designed as a product improvement — not a benchmark-specific fix — as it enables correct temporal grounding for any retrospective ingestion (meeting transcripts, email archives, historical documents).

4.2 Decomposer Extraction Depth

Early development revealed that the default decomposer prompt was too conservative. Server logs showed 18–41 atoms produced per session but only 4–6 surviving into storage; analysis of incorrect benchmark answers confirmed that facts present in the source conversation were absent from the atom store. Three changes addressed this:

1. Extract *all* distinct facts, opinions, reactions, and minor details, erring on the side of over-extraction (targeting 15–30 atoms per session)
2. Strengthen temporal resolution: every episodic atom *must* include when it happened, and all relative dates *must* resolve to absolutes against the provided `remembered_on` context
3. Raise `max_tokens` from 2,048 to 4,096 to prevent silent JSON truncation that was dropping entire session outputs on longer conversations

These changes were developed alongside the deduplication threshold adjustment described below, and shipped together.

4.3 Proper Noun Preservation

A separate failure pattern emerged from answer analysis: the decomposer was generalising away proper nouns — "Under Armour" becoming "a renowned outdoor gear company", "Exploding Kittens" becoming "a card game about cats". This discarded the specific information the benchmark tests. An explicit rule was added: brand names, product names, book titles, shop names, place names, and all other named entities must be preserved exactly as stated. The open-domain category — which frequently requires recall of specific named entities — improved by 18.7 percentage points following this change.

Table 1: LoCoMo benchmark results with gte-small embeddings (LLM-as-Judge accuracy, %). Mnemo uses Claude Sonnet 4.6 as both answer model and judge. Published baselines use varying evaluation configurations.

System	Single-Hop	Multi-Hop	Open-Domain	Temporal	Overall
Backboard	89.4	75.0	91.2	91.9	90.0
Hindsight (Gemini-3)	86.2	70.8	95.1	83.8	89.6
Hindsight (OSS-120B)	76.8	62.5	93.7	79.4	85.7
Hindsight (OSS-20B)	74.1	64.6	91.0	76.3	83.2
Mnemo (gte-small)	**78.7**	**74.5**	**70.8**	**76.5**	**76.1**
Memobase	70.9	46.9	77.2	85.1	75.8
Zep/Graphiti	74.1	66.0	67.7	79.8	75.1
Mem0-Graph	65.7	47.2	75.7	58.1	68.4
Mem0	67.1	51.2	72.9	55.5	66.9
OpenAI Memory	63.8	42.9	62.3	21.7	52.9

Table 2: Atom retention under 0.97 dedup threshold for outlier conversations.

Conversation	Total Atoms Decomposed	Active After Dedup	Retention
Conv 1	382	87	22.8%
Conv 9	561	118	21.0%
Conv 0 (reference)	∼500	484	∼97%

4.4 Deduplication Threshold

The low atom survival rate had a second cause beyond prompt conservatism: at a cosine similarity threshold of 0.90, `gte-small`'s narrow effective score range (0.70–1.0) caused semantically distinct atoms to appear as duplicates. "Melanie went camping with her family" and "Melanie's kids enjoyed finding flowers while camping" embedded close enough to be merged, discarding the specific detail about flowers.

Raising the threshold from 0.90 to 0.97 increased atom survival from approximately 20% to 70–100% of decomposed atoms.

This finding confirms a long-standing architectural concern: `gte-small`'s poor score separation, with an effective dynamic range of approximately 0.70–1.0, is problematic for operations that depend on fine-grained similarity discrimination. The deduplication threshold and the embedding model's score distribution are coupled parameters that must be tuned together — a finding we believe is relevant to any memory system using embedding-based deduplication.

With these design decisions in place and the original `gte-small` (384d) embedding model, Mnemo reached 76.1% overall accuracy on the full 10-conversation LoCoMo benchmark (1,540 questions). Table 1 presents the per-category breakdown alongside published baselines.

Even with the narrow score separation of `gte-small`, Mnemo's multi-hop score of 74.5% is notable: it effectively equals Backboard (75.0%) and substantially outperforms all other systems including Hindsight's best configuration (Latimer et al. 2025) (70.8% with Gemini-3) and Zep (66.0%). Multi-hop questions require synthesising information across multiple conversation sessions — exactly the scenario where typed atoms with graph expansion provide structural advantage over flat-fact retrieval.

5 Deduplication Sensitivity Analysis

During the full benchmark run, two conversations exhibited markedly lower accuracy: conversation 1 (59.3%) and conversation 9 (44.9%), against an average of approximately 81% for the remaining eight. Investigation revealed that these conversations had dramatically lower atom retention after deduplication:

These conversations contained thematically overlapping content — speakers returning to the same topics across sessions — causing `gte-small` to embed distinct-but-related atoms within the 0.97 similarity radius.

Table 3: Deduplication threshold sensitivity for outlier conversations.

Configuration	Conv 1	Conv 9	Overall
Dedup @ 0.97	59.3%	44.9%	76.1%
Dedup @ 0.99	87.7%	∼80%	81.9%

Table 4: LoCoMo benchmark results with EmbeddingGemma (LLM-as-Judge accuracy, %). Mnemo uses Claude Sonnet 4.6 as both answer model and judge. Published baselines use varying evaluation configurations.

System	Single-Hop	Multi-Hop	Open-Domain	Temporal	Overall
Backboard	89.4	75.0	91.2	91.9	90.0
Hindsight (Gemini-3)	86.2	70.8	95.1	83.8	89.6
Hindsight (OSS-120B)	76.8	62.5	93.7	79.4	85.7
Hindsight (OSS-20B)	74.1	64.6	91.0	76.3	83.2
Mnemo (EmbeddingGemma)	**81.6**	**85.0**	**67.7**	**82.8**	**82.1**
Memobase	70.9	46.9	77.2	85.1	75.8
Zep/Graphiti	74.1	66.0	67.7	79.8	75.1
Mem0-Graph	65.7	47.2	75.7	58.1	68.4
Mem0	67.1	51.2	72.9	55.5	66.9
OpenAI Memory	63.8	42.9	62.3	21.7	52.9

Re-ingesting these two conversations with a relaxed threshold of 0.99 raised atom retention to 408 (conv 1) and 600+ (conv 9), and accuracy jumped to 87.7% and approximately 80% respectively. The recalculated overall accuracy was 81.9%.

This analysis demonstrates that the relationship between embedding model score separation and deduplication threshold is a critical, under-discussed parameter in memory system design. The performance ceiling with `gte-small` is limited by its narrow score distribution (effective range ~ 0.17) rather than by the retrieval or decomposition pipeline. This diagnosis motivated the embedding model migration described in the following section.

6 Embedding Model Migration

The deduplication sensitivity analysis identified embedding score separation as the primary bottleneck. A four-model comparison evaluated candidates against Mnemo's constraints: self-hosted on CPU (8,GB RAM), commercially permissive licence, and wider score distribution than `gte-small`'s effective range of $∼$0.17.

`EmbeddingGemma-300M` (Google DeepMind Team 2025) — a 308M-parameter model with 768-dimensional output and Matryoshka Representation Learning (MRL) support — achieved an effective score range of 0.33 (floor 0.19, top-1 similarity 0.36 on representative atom queries), nearly double `gte-small`'s usable signal. The wider separation means that semantically distinct atoms no longer cluster within the deduplication radius, eliminating the pathological atom loss observed in conversations 1 and 9.

The migration was operationally straightforward: a server configuration change, re-embedding of all atoms, and recalibration of the `min_similarity` threshold. Table 4 presents the benchmark results after the embedding model migration.

The embedding model migration lifted overall accuracy from 76.1% to 82.1%, with gains across all categories. The most striking improvement is multi-hop reasoning, which rose from 74.5% to 85.0% — the highest published score on the LoCoMo benchmark for this category, exceeding Backboard's 75.0% by 10 percentage points and Hindsight's best configuration (70.8% with Gemini-3) by 14 percentage points. This confirms that the typed-atom architecture with graph expansion provides a structural advantage for cross-session synthesis, and that this advantage was previously masked by the embedding model's inability to preserve distinct atoms through deduplication.

The comparison with Hindsight is instructive. Hindsight's strength lies overwhelmingly in open-domain questions, where its Gemini-3 configuration reaches 95.1% — far above Mnemo's 67.7%. This category rewards systems that combine retrieved memories with the answer model's world knowledge (recognising that "a card game about exploding cats" is *Exploding Kittens*, for example), which depends as much on

Table 5: Development trajectory across four full benchmark runs (1,540 questions, 10 conversations each). Dev runs 1–3 reflect iterative local development; changes were not always committed to version control prior to evaluation. Run 4 is the first run in which all active changes were committed before execution and constitutes the validated result.

Run	Configuration	Single-Hop	Multi-Hop	Open-Domain	Te
Dev run 1	Temporal grounding + dedup threshold (`gte-small`)	70.2	70.4	52.1	
Dev run 2	+ Proper noun preservation (`gte-small`)	78.7	74.5	70.8	
Dev run 3	+ `EmbeddingGemma-300M`	82.6	81.9	70.8	
Run 4 (validated)	+ ivfflat retrieval fixes	81.6	85.0	67.7	

the backbone LLM as on memory architecture. In contrast, multi-hop questions test the memory layer's ability to synthesise information distributed across sessions — the category where structural properties of the memory representation matter most. Mnemo and Hindsight occupy opposite ends of this trade-off: Hindsight's four-way parallel retrieval is optimised for breadth, while Mnemo's typed graph expansion is optimised for synthesis.

Table 5 summarises the development trajectory across all four benchmark runs.

Two retrieval-layer bugs were fixed in the commits preceding Run 4. First, a PostgreSQL `ORDER BY` alias in the `/recall` path prevented the ivfflat index from being used, causing every retrieval to fall back to a sequential scan across the full atom table. Second, the ivfflat `probes` parameter was left at its default of 1, meaning each query searched approximately 1% of the index. Setting `probes = 10` raised coverage to roughly 10%. These fixes are reflected in the multi-hop gain of +3.1pp between Dev run 3 and Run 4: multi-hop questions require cross-session synthesis across more atoms, so they are the category most sensitive to retrieval coverage. The slight regressions in single-hop (−1.0pp) and open-domain (−3.1pp) are consistent with changed retrieval dynamics after the embedding model and deduplication pipeline were fully committed together for the first time; the effects are plausible but cannot be individually isolated across these runs.

7 Memory Consolidation

Mnemo implements a periodic consolidation pipeline inspired by complementary learning systems theory in cognitive neuroscience, where episodic memories encoded in the hippocampus are gradually abstracted into semantic knowledge in the neocortex during sleep (McClelland, McNaughton, and O'Reilly 1995). In Mnemo's analogue, a background consolidation job runs on a configurable interval (default: 60 minutes), executing a five-step pipeline within PostgreSQL advisory-locked transactions.

7.1 Clustering and Generalisation

The core consolidation step identifies clusters of three or more episodic atoms belonging to the same agent that share overlapping `domain_tags` and pairwise embedding cosine similarity above 0.85. Connected components are identified via union-find over the qualifying pairs. For each cluster, the system creates a new semantic atom whose embedding is the L^2-normalised centroid of the cluster members' embeddings. The text content is not abstractively summarised; instead, the most confident member's text is preserved verbatim with a prefix indicating the cluster size (e.g., "Generalised from 5 observations: User made pour-over coffee on Monday"). Confidence is accumulated across the cluster: $\alpha_{\text{gen}} = \sum_i \alpha_i$ (evidence accumulation), $\beta_{\text{gen}} = \max_i \beta_i$ (uncertainty floor).

The generalised atom is linked to each cluster member via directed `generalises` edges (source: generalised atom, target: member). Source episodic atoms remain active — their provenance is preserved in the graph — but the generalised semantic atom provides a compact, high-confidence representation that can be retrieved without expanding the full cluster. Atoms already covered by a prior generalisation are skipped to prevent repeated consolidation of the same cluster.

Because centroid embeddings tend to be broadly similar to many queries — over-matching relative to their actual specificity — consolidated atoms receive a 15% ranking penalty during retrieval ($S_{\text{consolidated}} = S \times 0.85$; see Section 3.4). This ensures that specific, decomposer-produced atoms are preferred when both match a query.

7.2 Decay Reaping

Consolidation does not *apply* temporal decay — decay is computed continuously at query time by the `effective_confidence()` PostgreSQL function (see Section 3). Consolidation's role is to *reap* the results: atoms whose effective confidence has fallen below 0.05 are deactivated. This separation of concerns means that an atom's ranking degrades gradually with each retrieval, rather than dropping discontinuously at consolidation boundaries.

7.3 Duplicate Merging

The pipeline also detects active atom pairs (same agent, same type) with cosine similarity above 0.90 and merges them via Bayesian confidence update. The older atom is retained; the newer atom's evidence is absorbed ($\alpha_{\text{new}} = \alpha_{\text{old}} + \alpha_{\text{newer}} - 1$, clamped to ≥ 1), its edges are reassigned to the surviving atom, and it is deactivated. Merge events are recorded in the access log for auditability.

7.4 Edge Pruning and Agent Purging

Dead edges — those where either endpoint has been deactivated — are pruned to prevent graph expansion from traversing stale paths. Finally, agents whose `data_expires_at` timestamp has passed (set during departure, default retention: 30 days) are deleted, cascading to all associated atoms, views, edges, and snapshot atoms.

7.5 Consolidation and Benchmark Accuracy

The interaction between consolidation and benchmark accuracy has not yet been characterised. The LoCoMo evaluation ingests and queries within a single session, which does not exercise the consolidation pathway — no atoms have had time to decay, cluster, or be merged. Evaluating consolidation's impact on long-horizon agent performance over weeks or months of operation is an important direction for future work. Of particular interest is whether the specificity penalty adequately balances the trade-off between compact generalised atoms and the specific detail required for precise recall, and whether centroid-based embeddings remain effective as cluster size grows.

8 Addressing Common Concerns

We anticipate three questions that practitioners may raise about graph-based memory systems.

8.1 Temporal Edge Weighting and Belief Revision

A persistent concern with memory systems is how they handle contradictory information over time — for example, an agent that learns "I like coffee" and later encounters "I've quit caffeine."

Mnemo addresses this through Beta-distribution confidence rather than temporal decay. When contradicting evidence arrives, the existing atom's β parameter is incremented, reducing its expected confidence without deleting it. The original belief is preserved in the graph (enabling queries like "what did the user used to prefer?") while the newer, contradicting atom carries higher effective confidence and will be ranked above the older one in composite retrieval.

This is a principled alternative to the recency-bias approach used by most systems (where newer facts simply overwrite older ones). The Bayesian mechanism naturally handles partial contradictions, graduated certainty, and the distinction between "this was once true" and "this is currently true."

8.2 Entity Resolution

Mnemo does not currently perform explicit entity resolution — it does not automatically determine that "the boss," "Mr. Henderson," and "that guy from the meeting" refer to the same individual. Instead, the decomposer is instructed to preserve named entities exactly as stated, and the graph expansion mechanism during retrieval surfaces related atoms through typed edges.

In practice, this means that a recall query for "Mr. Henderson" will retrieve atoms mentioning him by name, and graph expansion may surface atoms about "the boss" if they are linked through co-occurrence edges

inferred during decomposition. Full entity resolution with coreference tracking is a planned enhancement, but we note that the current approach — preserving specificity and relying on graph expansion — performs well on the LoCoMo benchmark without explicit entity merging.

8.3 Global Context and Cluster Summarisation

Can the graph summarise the "vibe" of a large atom cluster without reading every edge? Mnemo's view mechanism provides a partial answer: a view can filter atoms by type, domain tags, or confidence threshold, producing a scoped projection of the knowledge graph. The `export_skill` endpoint can render a view as a structured summary suitable for injection into an agent's system prompt.

For true cluster-level summarisation — distilling hundreds of atoms into a coherent narrative — the consolidation pipeline (Section 7) provides the infrastructure. Its current implementation creates generalised semantic atoms from episodic clusters using centroid embeddings and the most confident member's text. More sophisticated abstractive summarisation — using an LLM to synthesise cluster content into a natural-language summary rather than selecting a representative member — is a planned enhancement. The consolidation architecture supports this extension without structural changes; only the text-generation step within `_cluster_and_generalise` would need to be replaced.

9 Limitations and Future Work

Embedding model generalisation. The migration from `gte-small` to `EmbeddingGemma-300M` resolved the immediate score-separation bottleneck, but the interaction between embedding model, deduplication threshold, and content domain remains a coupled parameter space. Whether EmbeddingGemma's wider score distribution generalises to domains beyond conversational memory (e.g., technical documentation, code, multilingual content) is not yet characterised. The MRL capability offers a potential mitigation — truncating to 512 or 256 dimensions for specific domains — but this has not been benchmarked.

Open-domain questions. Mnemo's weakest category (67.7%) includes questions requiring world knowledge inference — determining that a "card game about exploding cats" is *Exploding Kittens*, or that Lake Tahoe borders California and Nevada. These questions conflate memory recall with general knowledge application. Addressing this requires either more aggressive proper-noun preservation in the decomposer or allowing the answer model to supplement recalled context with its own knowledge, which changes the evaluation semantics.

Benchmark methodology. LoCoMo does not have a standardised leaderboard — published results use varying judge models, question subsets, and evaluation prompts. Our results use Claude Sonnet 4.6 as both answer model and judge. Direct comparison with systems evaluated using GPT-4o-mini or GPT-4.1 requires caution. We plan to publish results with multiple judge configurations for full comparability.

Sharing evaluation. No existing benchmark evaluates agent-to-agent memory sharing. Mnemo's sharing mechanism — capability-controlled views with typed atoms — is architecturally novel but empirically unvalidated at scale. Designing a multi-agent memory benchmark is an important direction for future work.

Consolidation evaluation. The consolidation pipeline has not been evaluated under long-horizon workloads. Key open questions include: whether centroid embeddings degrade in quality as cluster size grows, whether the 15% specificity penalty is the right magnitude across domains, and whether the current text-selection strategy (most confident member) should be replaced with LLM-generated abstractive summaries.

Information-geometric extensions. The Beta-distribution confidence model admits a natural geometric interpretation. The space of atom confidences $\mathrm{Beta}(\alpha, \beta)$ forms a two-dimensional statistical manifold equipped with the Fisher information metric (Amari 2016), where geodesic distance between confidence states corresponds to the minimum information cost of transitioning between them. This structure suggests several directions: natural-gradient confidence updates that respect the intrinsic geometry of the parameter space rather than operating in arbitrary coordinates; α-projections for principled consolidation, minimising information loss when merging episodic clusters into semantic atoms; and Fisher–Rao distance as a geometrically grounded signal for contradiction detection. These extensions are theoretically motivated but empirically unvalidated. Whether natural-gradient updates produce measurably better calibration than coordinate-space increments on realistic agent workloads is an open question we intend to pursue using Mnemo's existing benchmark infrastructure.

10 Conclusion

We have introduced Agent Experience (AX) as a design discipline for persistent AI agents, with memory architecture as its central concern. Mnemo — a typed-atom memory system with Bayesian confidence, composite retrieval, and agent-to-agent sharing — achieves 82.1% on the LoCoMo benchmark, placing second overall and first in multi-hop reasoning at 85.0%, surpassing all published baselines for cross-session synthesis.

The diagnostic trajectory from 69.5% to 82.1% — achieved through temporal grounding, deduplication tuning, decomposer specificity, and an embedding model migration from `gte-small` to `EmbeddingGemma-300M` — demonstrates that memory system performance depends on a chain of interacting design decisions, each of which can be systematically diagnosed and improved. The deduplication sensitivity analysis identified embedding score separation as the primary bottleneck, and the subsequent migration confirmed the hypothesis: wider score distribution eliminated pathological atom loss and unlocked the full potential of the typed-atom architecture.

More broadly, we argue that as agents become persistent collaborators — accumulating knowledge, forming beliefs, and working in teams — the principles governing their memory deserve the same rigour that UX brings to human interfaces. The quality of an agent's memory is not a feature; it is the foundation of its experience.

Mnemo is available at `http://github.com/inforge-ai/mnemo-server`. The benchmark harness and evaluation code will be released alongside this paper, and can be found at `https://github.com/inforge-ai/locomo_bench`.

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